AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395
Applied Cybersecurity

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 10/Apr./2025

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Number of Exam Pages: 3	**Time Allowed:** 90 m

Instructions:

- Total Marks: 30
- Total Number of Questions:3
- Marks Distribution:
  - Question 1: 10 marks
  - Question 2: 10 marks
  - Question 3: 10 marks
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted
- Be sure you write your name and ID on the External Answer booklet.

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Question 1: Select the best answer : **Total: 10 Marks**

1. Which of the following usually observe each activity on the internet of the victim, gather all information in the background, and send it to someone else?
   a. Malware
   b. Spyware
   c. Adware
   d. All of the above
2. Which of the following ensures that only authorized parties can view protected information? (1 Mark)
   a. Authorization
   b. Confidentiality
   c. Availability
   d. Integrity
3. identifying users' needs. Which of these generally recognized security positions has Ian been offered?(1 Mark)
   a. Security administrator
   b. Security technician
   c. Security officer
   d. Security manager
4. A(n) _____ is a program that secretly attaches itself to a carrier such as a document or program and then executes when that document is opened or program is launched. (1 Mark)
   a. virus
   b. worm
   c. rootkit
   d. Trojan
5. The first action that a virus takes once it infects a computer is to(1 Mark)
   a. close all ports
   b. erase the payload
   c. authenticate
   d. replicate
6. A(n) _____ works on the principle of comparing new behavior against normal behavior.
   a. Host Intrusion Detection System (HIDS) (1 Mark)
   b. packet filter
   c. Internet Resource Chat (IRC)
   d. personal software firewall
7. A computer program that pretends to clean up a hard drive but actually performs a malicious activity is known as a(n) (1 Mark)
   a. Trojan
   b. Rootkit
   c. worm
   d. logic bomb
8. A(n) _____ is either a small hardware device or a program that monitors each keystroke a user types on the computer's keyboard. (1 Mark)
   a. keylogger
   b. macro
   c. script kiddie
   d. port replicator

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9. The _____ is the software that runs on a physical computer and manages multiple virtual machine operating systems. (1 Mark)
    a. virtual resource allocator (VRA)
    b. hypervisor
    c. Microsoft Control Plug-in
    d. hardware allocator
10. _____ is exploiting a vulnerability in the software to gain access to resources that the user would normally be restricted from obtaining. (1 Mark)
    a. Vulnerability assessment
    b. Software threat release (STR)
    c. Hardware virus
    d. Privilege escalation

Question 2: Describe the following general kinds of attacks, in a few sentences: Total: 10 Marks

1. How do virtualization and virtual machine sprawl present both opportunities and challenges for network security?(2 Marks)
2. How does a demilitarized zone (DMZ) enhance the security of a network? (2 Marks)
3. Discuss the importance of Network Access Control (NAC) in maintaining a secure network environment. (2 Marks)
4. How do SQL injection attacks work, and what are some examples of their impact? (2 Marks)
5. What are Denial of Service (DoS) attacks, and what are some common types? (2 Marks)

Question 3 Answer the following questions **[10 Marks]**

1. What are some of the challenges in defending against attacks? (2 Marks)
2. What are the primary traits by which malware is classified? (2 Marks)
3. What are interception attacks, and can you describe the main types? (2 Marks)
4. Describe the concept of privilege escalation and differentiate between its types. (2 Marks)
5. How do load balancers contribute to network security and performance? (2 Marks)

End of questions

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